



CENTRALISED COLLEGE DATABASE MANAGEMENT SYSTEM



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INTRODUCTION TO DBMS AND CENTRALIZED DATABASE

- A Database Management System (DBMS) is software used to store, manage, and retrieve data efficiently.
- In colleges, large amounts of data such as student records, faculty details, courses, attendance, and results need to be handled.
- A centralized database system stores all data in one place, making it easier to manage and access information quickly and accurately.

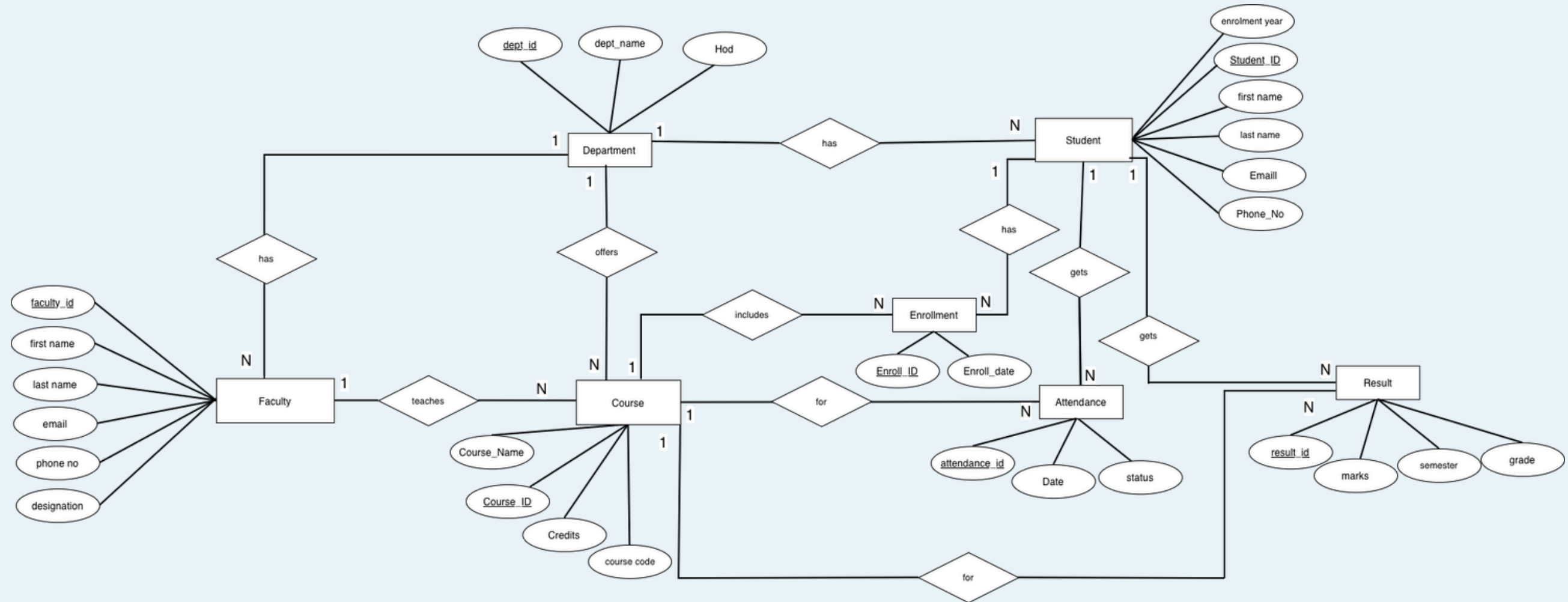
3. PROJECT BACKGROUND

In many colleges, information is stored in different files or systems, which creates several problems:

- Data redundancy and duplication
- Difficulty in managing student and faculty information
- Time-consuming report generation
- Risk of data inconsistency

A centralized system helps integrate all academic information into a single database.

ER DIAGRAM



DATABASE TABLES

The following tables are designed to organize college data efficiently and maintain relationships between entities.

Table	Primary Key	Key Attributes
Student	StudentID	Name, DOB, Email, DeptID
Faculty	FacultyID	Name, Designation, DeptID
Department	DeptID	DeptName, HOD
Course	CourseID	CourseName, Credits, FacultyID
Enrollment	EnrollmentID	StudentID, CourseID, Date
Attendance	AttendanceID	StudentID, CourseID, Status
Result	ResultID	StudentID, CourseID, Marks, Grade



SQL QUERIES USED

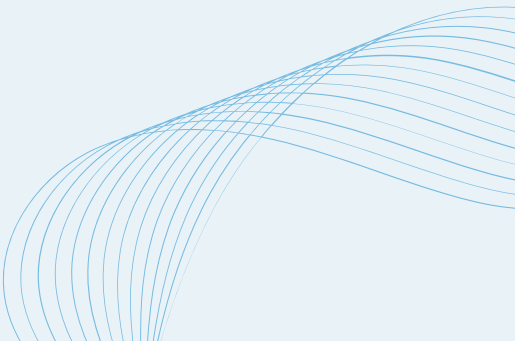
The system uses SQL queries to retrieve and manage data efficiently from multiple related tables.

JOIN Queries-
It combines data from multiple tables to display meaningful information such as student details with course and department.

Aggregate functions –
It perform calculations like average marks, total students, and count of enrollments.

Subqueries –
It help filter data based on conditions from another query.

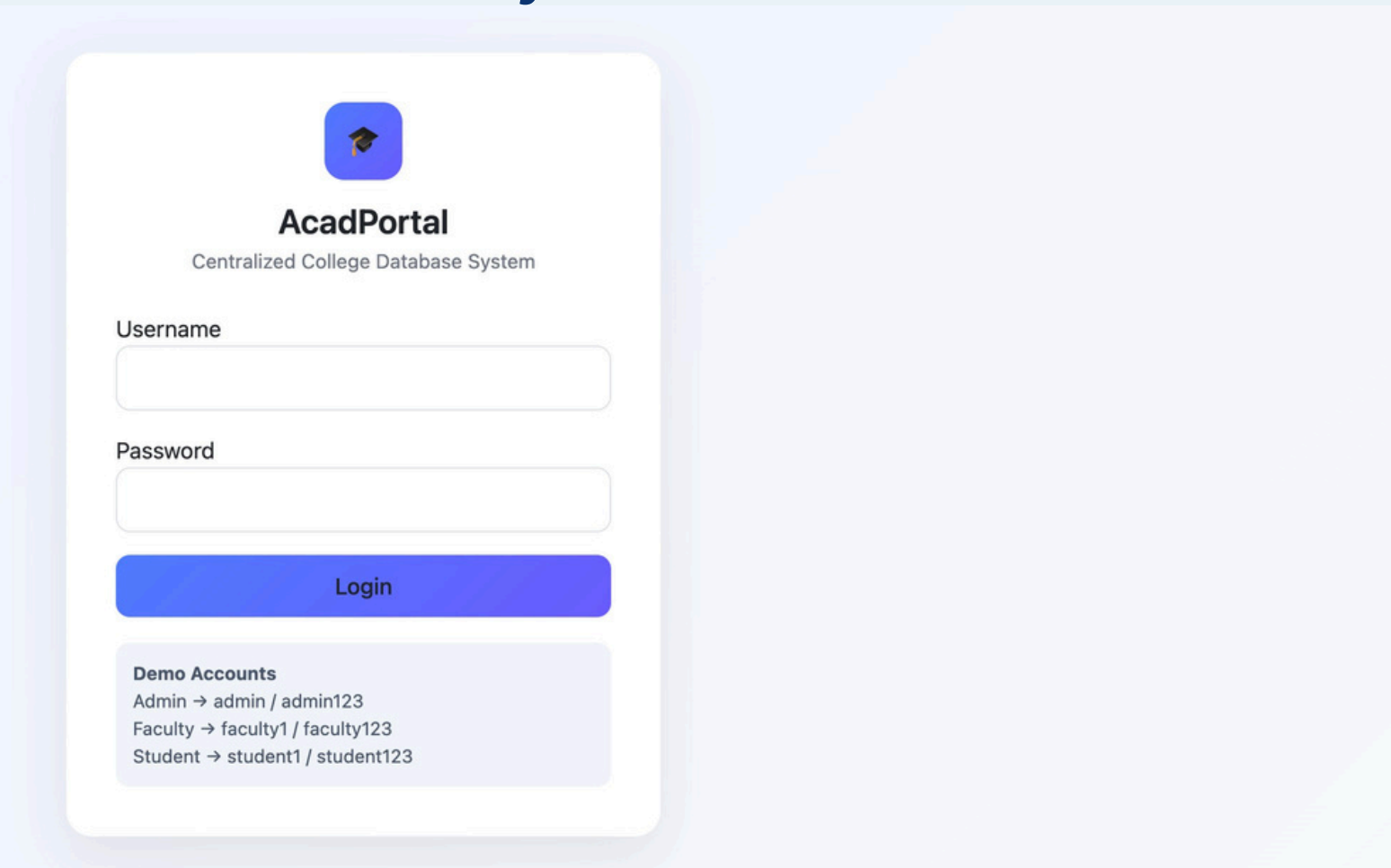
Views –
store important queries so data can be accessed quickly without writing complex SQL again.



USER INTERFACE SCREENS

Login Page:

- Secure login for Admin, Faculty, and Students.
- Users enter username and password to access the system



The image shows a mockup of the AcadPortal login page. It features a white login card with rounded corners and a subtle shadow, centered on a light blue background with abstract wavy lines. At the top of the card is a blue square icon with a white graduation cap. Below the icon, the text 'AcadPortal' is displayed in a bold, black font, followed by 'Centralized College Database System' in a smaller, gray font. The login form consists of two white input fields with thin gray borders, labeled 'Username' and 'Password' in a small, gray font. Below the password field is a prominent blue 'Login' button with white text. At the bottom of the card, a light purple box contains the heading 'Demo Accounts' and three lines of text: 'Admin → admin / admin123', 'Faculty → faculty1 / faculty123', and 'Student → student1 / student123'.



AcadPortal
Centralized College Database System

Username

Password

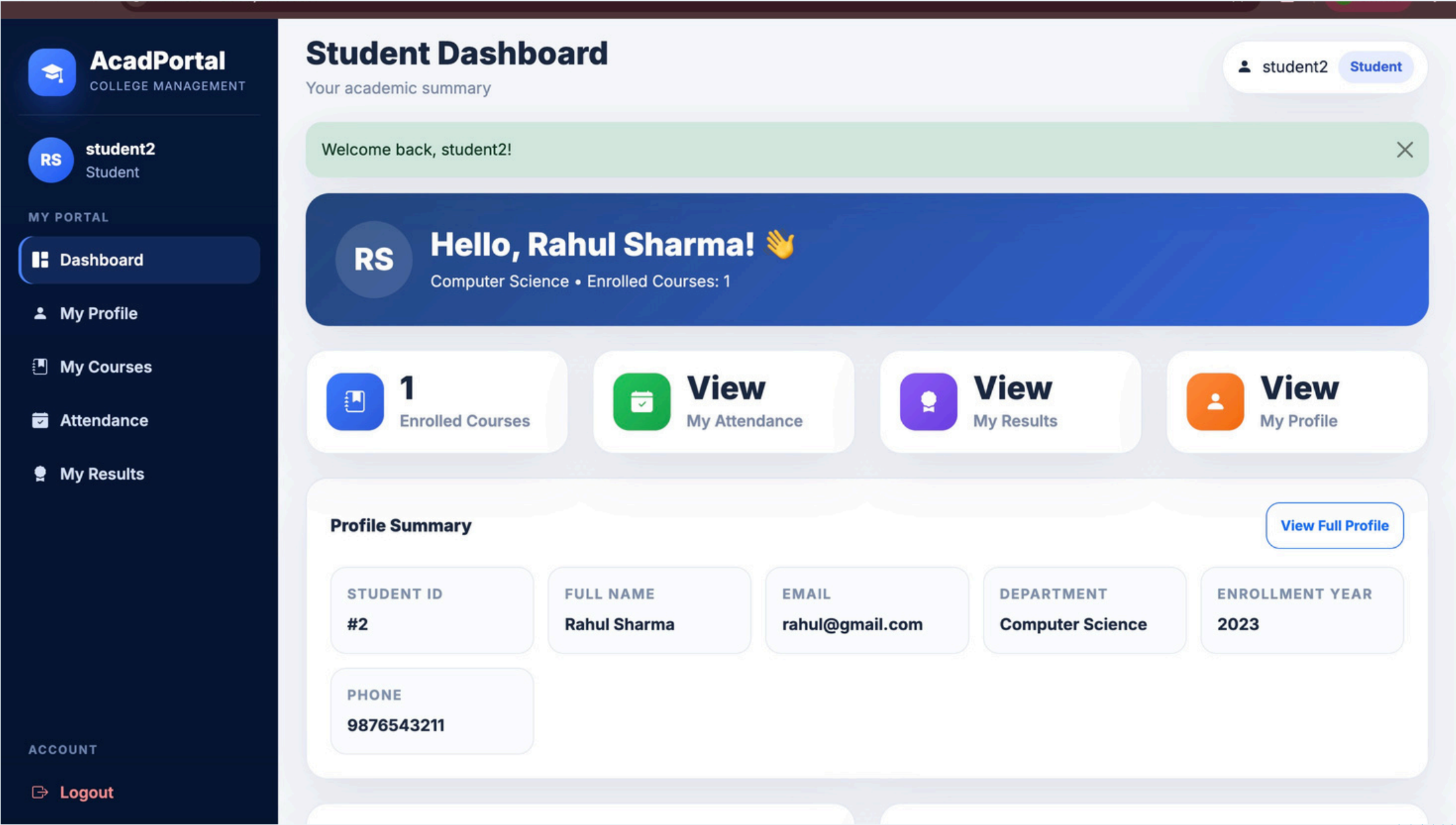
Login

Demo Accounts
Admin → admin / admin123
Faculty → faculty1 / faculty123
Student → student1 / student123

Student Dashboard

Displays student academic information.

- View courses
- Check attendance
- View results
- Profile details



Admin Dashboard

Central control panel for managing data.

- **Manage students**
- **Manage faculty**

- **Manage courses**
- **Manage departments**



AcadPortal
COLLEGE MANAGEMENT



admin
Admin

MY PORTAL

Dashboard
My Profile
Students
Add Student
Faculty
Add Faculty
Courses
Attendance
Results

ACCOUNT
Logout

Admin Dashboard
Overview of students, faculty, courses, and recent activity

admin
Admin

Welcome back, admin!

ADMINISTRATION CENTER
Hello, Admin 🖐️

Add Student
Add Faculty

7
Students

4
Faculty

5
Courses

5
Departments

Department Snapshot
Live Data

DEPARTMENT	STUDENTS	FACULTY
biology	0	0
Civil	1	0
Computer Science	4	2
Electronics	1	1

Quick Actions

View all students
View all faculty
View courses
Check results

Faculty Dashboard

Helps faculty manage academic activities.

- View assigned courses
- Monitor student performance
- Manage attendance

 **AcadPortal**
COLLEGE MANAGEMENT

RK

faculty1
Faculty

MY PORTAL

Dashboard

My Profile

Faculty Directory

My Courses

Attendance

Results

ACCOUNT

Logout

Faculty Dashboard

Manage your courses and monitor student academic data

faculty1 Faculty

Logged out successfully.

Welcome back, faculty1!

FACULTY WORKSPACE

Ramesh Kumar

Professor • Computer Science

1

Courses

3

Students

Profile

NAME

Ramesh Kumar

EMAIL

ramesh@college.com

PHONE

9876543201

Assigned Courses

CODE	COURSE	CREDITS	STUDENTS
DBMS101	Database Management System	4	3

CONCLUSION

The Centralized College Database Management System helps manage academic information efficiently.

It reduces data redundancy, improves accuracy, and makes data retrieval faster.

The system provides a structured way to handle student, faculty, course, attendance, and result data.

This project demonstrates the practical use of DBMS concepts in real-world applications.



THANK YOU